THE RESIDUE FIELD OF THE HENSELIAN LOCAL RING EQUALS THE RESIDUE FIELD AT THE POINT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2793-2803, immediately following the statement at 2783-2791.

Editorial note. The parenthetical at source lines 2800-2802 (the lemma's output is the construction/definition of the henselization) is part of the counted proof and is the reason the final step is a reading, not a new argument.

Editorial note. One cross-chapter pointer pair (Algebra lemma + definition) and one same-chapter stub are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Henselian local ring via elementary etale neighbourhoods. Lemma lemma-describe-henselian-local-ring of this chapter, cited at proof lines 2797-2798. O^h_{X,x} = O^h_{U,u} for the chosen chart – the bridge from X to the scheme chart.
- Residue field is etale local. chart property invoked at proof lines 2795-2796: kappa(u) = kappa(x) for an elementary etale neighbourhood. Aligns the two residue fields before the henselization computation.
- Henselization does not change the residue field. Algebra, Lemma algebra-lemma-henselization with Definition algebra-definition-henselization, cited at proof lines 2799-2803. The residue field of O^h_{U,u} is kappa(u), since henselization is built as a colimit of etale neighbourhoods of the local ring.

Source reference. Original reference 04GQ: focused-seeds/geometry/S9-decent-residue-field-henselian-local-ring/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04GN: focused-seeds/geometry/S9-decent-residue-field-henselian-local-ring/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 0BGW: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2716–2716. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 0EMW: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2597–2597. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0EMY: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2732–2732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $x \in |X|$. The residue field of the henselian local ring of X at x (Definition 0BGW) is the residue field of X at x (Definition 0EMW).*

Proof. Choose an elementary étale neighbourhood $(U, u) \to (X, x)$. Then $\kappa(u) = \kappa(x)$ and $\mathcal{O}^h_{X,x} = \mathcal{O}^h_{U,u}$ (Lemma 0EMY). The residue field of $\mathcal{O}^h_{U,u}$ is $\kappa(u)$ by Algebra, Lemma 04GN (the output of this lemma is the construction/definition of the henselization of a local ring, see Algebra, Definition 04GQ). □